

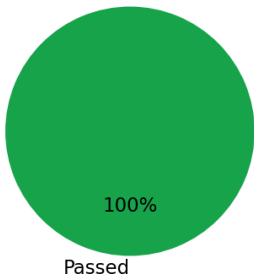
Bayes Tree Engine

Validation Report

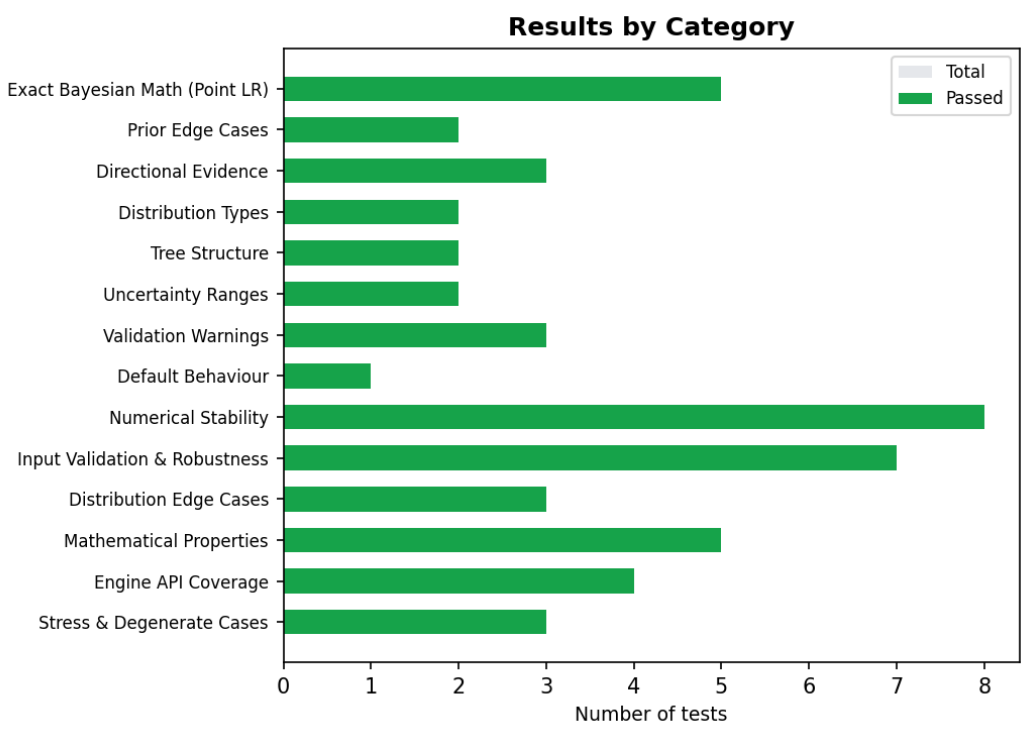
Tests: 50 | Passed: 50 | Failed: 0

Simulations per test: 10,000
Generated: 2026-05-08 15:32

50/50 Tests Passed



Results by Category



Detailed Test Results

Exact Bayesian Math (Point LR)

#	Test	Result	Details
01	Point LR — single for	PASS	median=0.6667, expected=0.6667
02	Point LR — single against	PASS	median=0.3333, expected=0.3333
03	Point LR — cancelling evidence	PASS	median=0.5000, expected=0.5000
04	Point LR — combined for	PASS	median=0.8000, expected=0.8000
05	Point LR — three mixed	PASS	median=0.7500, expected=0.7500

Prior Edge Cases

#	Test	Result	Details
06	Prior edge — very low (0.01)	PASS	median=0.0917, expected=0.0917
07	Prior edge — very high (0.99)	PASS	median=0.9083, expected=0.9083

Directional Evidence

#	Test	Result	Details
08	All evidence FOR	PASS	median=0.9712 > 0.85
09	All evidence AGAINST	PASS	median=0.0005 < 0.01
10	Neutral only — posterior $\approx$ prior	PASS	median=0.4950 $\approx$ 0.50

Distribution Types

#	Test	Result	Details
11	Distribution — uniform	PASS	median=0.4330, std=0.1180
12	Distribution — beta	PASS	median=0.5927, std=0.0713

Tree Structure

#	Test	Result	Details
13	Structure — deep nesting (4 levels)	PASS	median=0.4020, 10000 samples
14	Structure — many children (8)	PASS	median=0.2320, 10000 samples

Uncertainty Ranges

#	Test	Result	Details
15	Edge — wide LR uncertainty	PASS	90% CI spread=0.9898 > 0.30
16	Edge — narrow LR (near-deterministic)	PASS	median=0.5000, std=0.002970

Validation Warnings

#	Test	Result	Details
17	Warning — type vs LR conflict	PASS	2 warnings (expected 2)
18	Warning — $lr_{min} > lr_{max}$	PASS	1 order warning(s)

#	Test	Result	Details
19	Warning — $lr_min \leq 0$	PASS	1 zero/negative warning(s)

Default Behaviour

#	Test	Result	Details
20	Default prior (none specified)	PASS	median=0.6667, expected=0.6667

Numerical Stability

#	Test	Result	Details
21	Numerical — prior=0 and prior=1	PASS	prior=0→0.000000, prior=1→1.000000
22	Numerical — LR=0 and LR=-1	PASS	median=0.000000, no crash
23	Numerical — extreme LR $1e\pm300$	PASS	LR=1e-300→1.00e-12, LR=1e+300→1.0000000000
24	Numerical — LR=1 identity	PASS	median=0.3000, expected=0.3000
25	Numerical — to_lo/from_lo round-trip	PASS	max error=1.11e-16
26	Numerical — post_to_lr round-trip	PASS	max rel error=3.55e-16
27	Numerical — from_lo clamp $\pm700/\pm1000$	PASS	from_lo(700)=1.0000000000, from_lo(-700)=9.86e-305
28	Numerical — accumulation overflow	PASS	overflow→1.000000, underflow→1.00e-40

Input Validation & Robustness

#	Test	Result	Details
29	Input — empty children list	PASS	median=0.3500, expected=0.3500
30	Input — prior out of [0,1]	PASS	prior=1.5→1.0000, prior=-0.5→0.000000
31	Input — $lr_min == lr_max$ degenerate	PASS	median=0.6667, std=0.000000
32	Input — partial range spec	PASS	lr_min only→0.6339, lr_max only→0.3542
33	Input — unknown lr_dist	PASS	median=0.7093, std=0.0578
34	Input — point LR precedence	PASS	median=0.6667, std=0.000000
35	Input — nested validation warning	PASS	1 nested warning(s)

Distribution Edge Cases

#	Test	Result	Details
36	Dist — beta(1,1) $\approx$ uniform	PASS	diff=0.0016 < 0.03
37	Dist — beta extreme params	PASS	median=0.4248, std=0.1111
38	Dist — log_uniform near-zero clamp	PASS	median=0.0000

Mathematical Properties

#	Test	Result	Details
39	Math — monotonicity	PASS	LR mono=True, prior mono=True
40	Math — symmetry around 0.5	PASS	max error=1.11e-16
41	Math — commutativity	PASS	diff=0.0010 < 0.02
42	Math — log-odds additivity $2\times3=9$	PASS	$2\times LR=3=0.9000$, $1\times LR=9=0.9000$
43	Math — neutral identity (LR=1)	PASS	base=0.6667, +LR=1=0.6667

Engine API Coverage

#	Test	Result	Details
44	API — run_simulation full pipeline	PASS	keys OK, median=0.6667
45	API — progress_callback	PASS	5 calls, final reached
46	API — collect() NodeResult tree	PASS	2 children, median=0.4011
47	API — importance ranking	PASS	8 items, sorted correctly

Stress & Degenerate Cases

#	Test	Result	Details
48	Stress — 50 branches	PASS	median=1.0000, 50 branches OK
49	Stress — single-child LOO	PASS	without=0.5000 $\approx$ prior
50	Stress — all point LR's zero variance	PASS	std=0.00e+00

Test Coverage Summary

- **Core Bayesian math** — exact posterior verification with point LRs, cancellation, combination, and three-branch mixed evidence.
- **Prior handling** — extreme priors (0.01, 0.99), boundary priors (0, 1), out-of-range priors (1.5, -0.5), and default prior omission.
- **Distributions** — uniform, log_uniform, beta with standard and extreme parameters (U-shaped, concentrated, alpha=beta=1 equivalence).
- **Numerical stability** — LR=0, LR=-1, LR=1e±300, to_lo/from_lo round-trips, from_lo clamp at ±700/±1000, and accumulation overflow/underflow with 20 extreme children.
- **Input robustness** — empty children, degenerate lr_min==lr_max, partial range specification, unknown lr_dist fallback, point LR precedence over ranges.
- **Validation warnings** — evidence_type vs LR conflicts, inverted lr_min/lr_max, lr_min≤0, and warnings detected in deeply nested children.
- **Mathematical properties** — monotonicity (LR and prior), symmetry around 0.5, commutativity of children order, log-odds additivity ($2 \times \text{LR} = 3 \equiv \text{LR} = 9$), neutral identity (LR=1 no-op).
- **Engine API** — run_simulation full pipeline, progress_callback invocation, collect()/NodeResult tree structure, importance ranking sort order.
- **Stress tests** — 50-branch tree, single-child leave-one-out, all-point-LR zero variance.

Report generated by validation suite on 2026-05-08 15:32. All 50 tests executed with 10,000 Monte Carlo simulations each (seed=42 for reproducibility).